

12.536

EE25BTECH11002 - Achat Parth Kalpesh

Question:

The rank of the matrix $\begin{pmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{pmatrix}$ is

1) 1

2) 2

3) 3

4) 4

Solution:

We perform row operations to reduce it to row echelon form:

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 4 & 1 & 3 & 1 \end{pmatrix} \quad (4.1)$$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 4 & 1 & 3 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 4R_1} \begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & -3 & 3 & 9 \end{pmatrix} \quad (4.2)$$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & -2 & 2 & 6 \\ 0 & -3 & 3 & 9 \end{pmatrix} \xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \quad (4.3)$$

$$\begin{pmatrix} 1 & 1 & 0 & -2 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \quad (4.4)$$

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & -3 & 3 & 9 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + 3R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (4.5)$$

Now the matrix is in reduced row echelon form

The number of non-zero rows = 2

$$\text{rank}(A) = 2 \quad (4.6)$$